

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collects and processes historical weather and rainfall data for training and testing.	Data Preprocessing	Done	10
2	Data Preprocessing	Cleans missing values, prepares features, and formats the dataset for machine learning.	Data Processing	Done	10
3	Machine Learning Model Training	Trains Decision Tree, Random Forest, KNN, and XGBoost models using historical flood data.	ML Model	Done	20
4	Flood Prediction	Predicts whether a flood is likely based on rainfall, cloud visibility, and seasonal rainfall inputs.	Prediction Engine	Done	20
5	Model Evaluation	Compares model performance and selects the best-performing model (XGBoost – 96.55% accuracy).	Evaluation Module	Done	15
6	Flask Web Application	Provides a user-friendly web interface for entering weather data and viewing predictions.	Web Application	Done	15
7	Cloud Deployment Support	Supports deployment on IBM Cloud for remote accessibility and scalability.	Deployment Module	Done	5
8	Disaster Management Support	Assists authorities in early warning, evacuation planning, and resource allocation based on predictions.	Decision Support System	Done	5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Flask Web App Dashboard
2	Backend / Logic	3	Data Processing, ML Model, Prediction Engine
3	Database / Storage	1	Historical Weather Dataset Storage
4	API / Integration	1	Weather Data API (Rainfall, Temperature)
5	Security / Authentication	1	Basic User Access / Input Validation